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A computational model of chemically and mechanically induced thrombus formation

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LadHyX

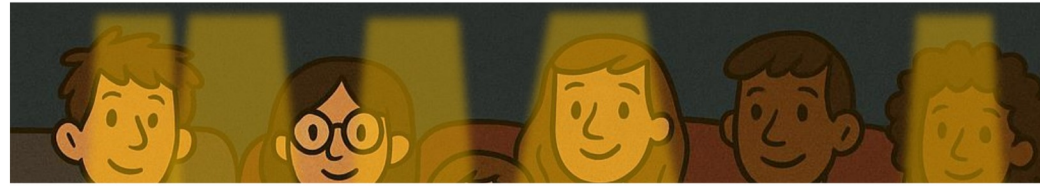


AXA
Research Fund



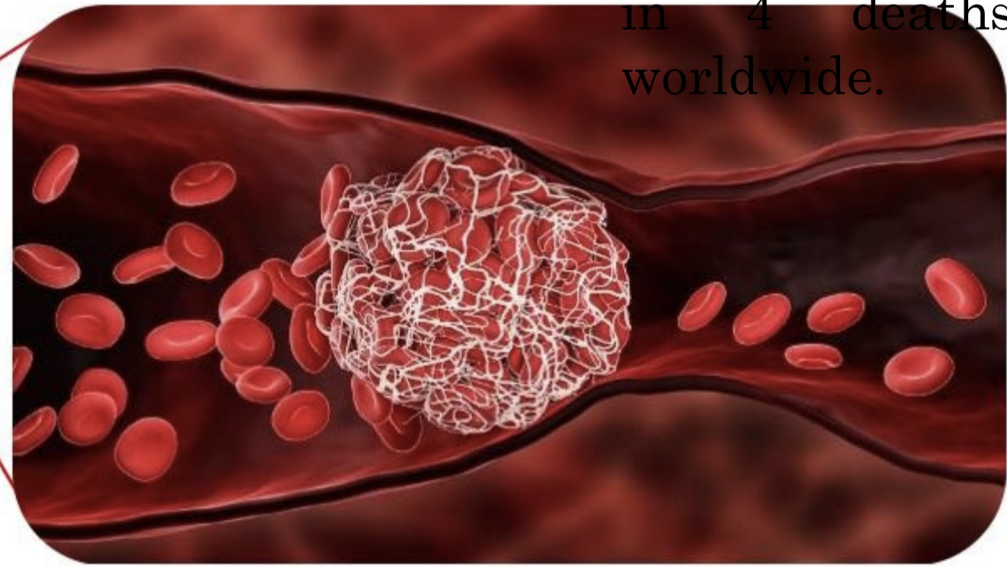
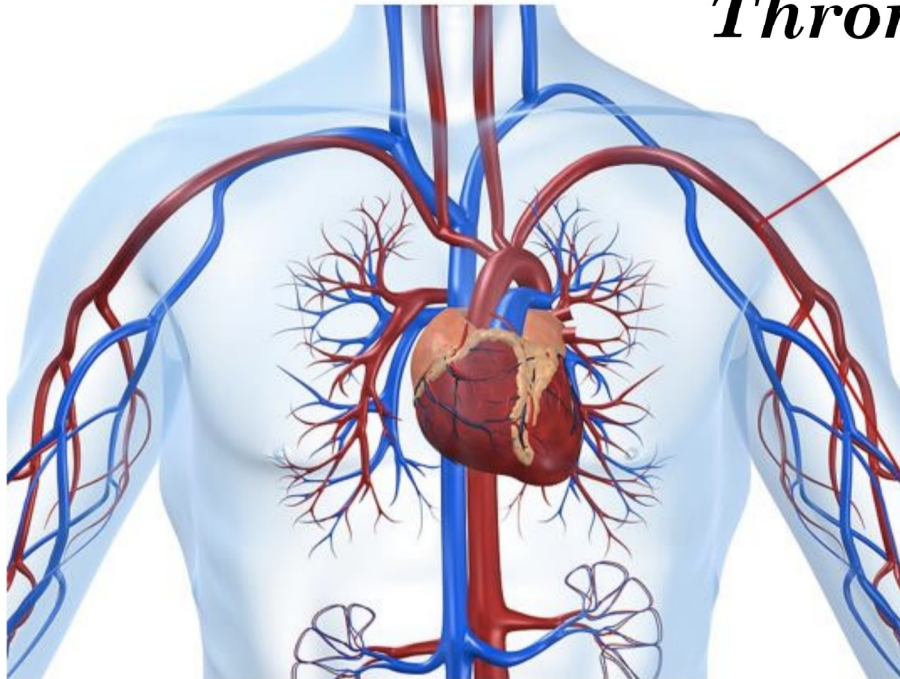
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Introduction: Motivation behind our work



Thrombotic conditions have been estimated to account for 1 in 4 deaths worldwide.

Thrombus formation

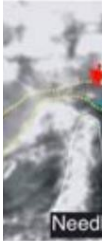


Introduction: Motivation behind our work

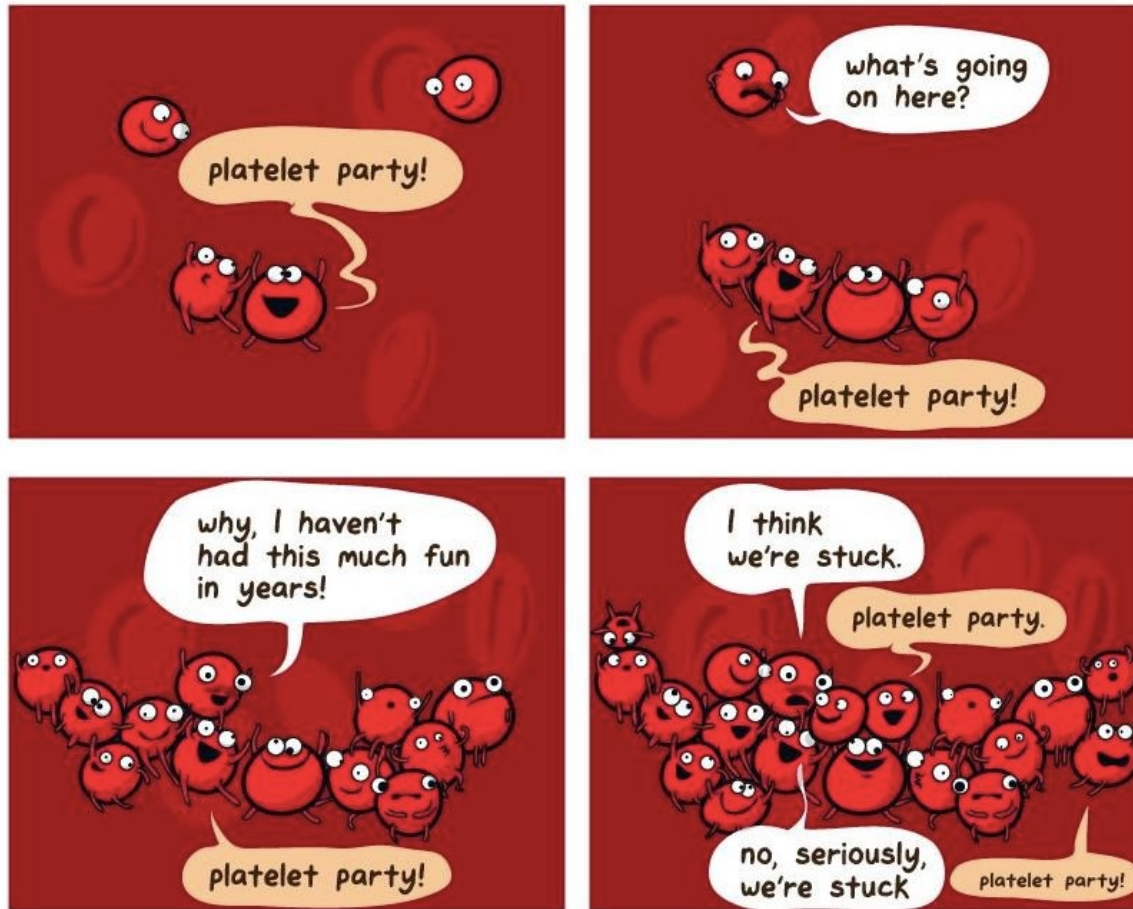
New insight:

It has been experimentally driven

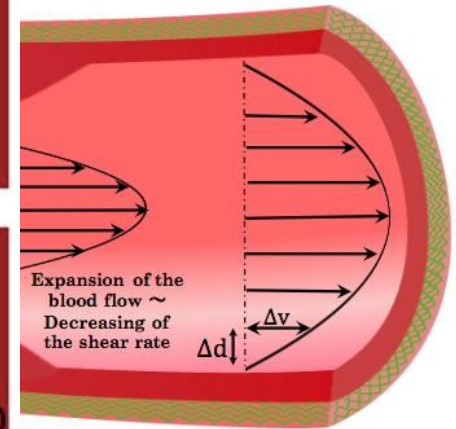
In vivo



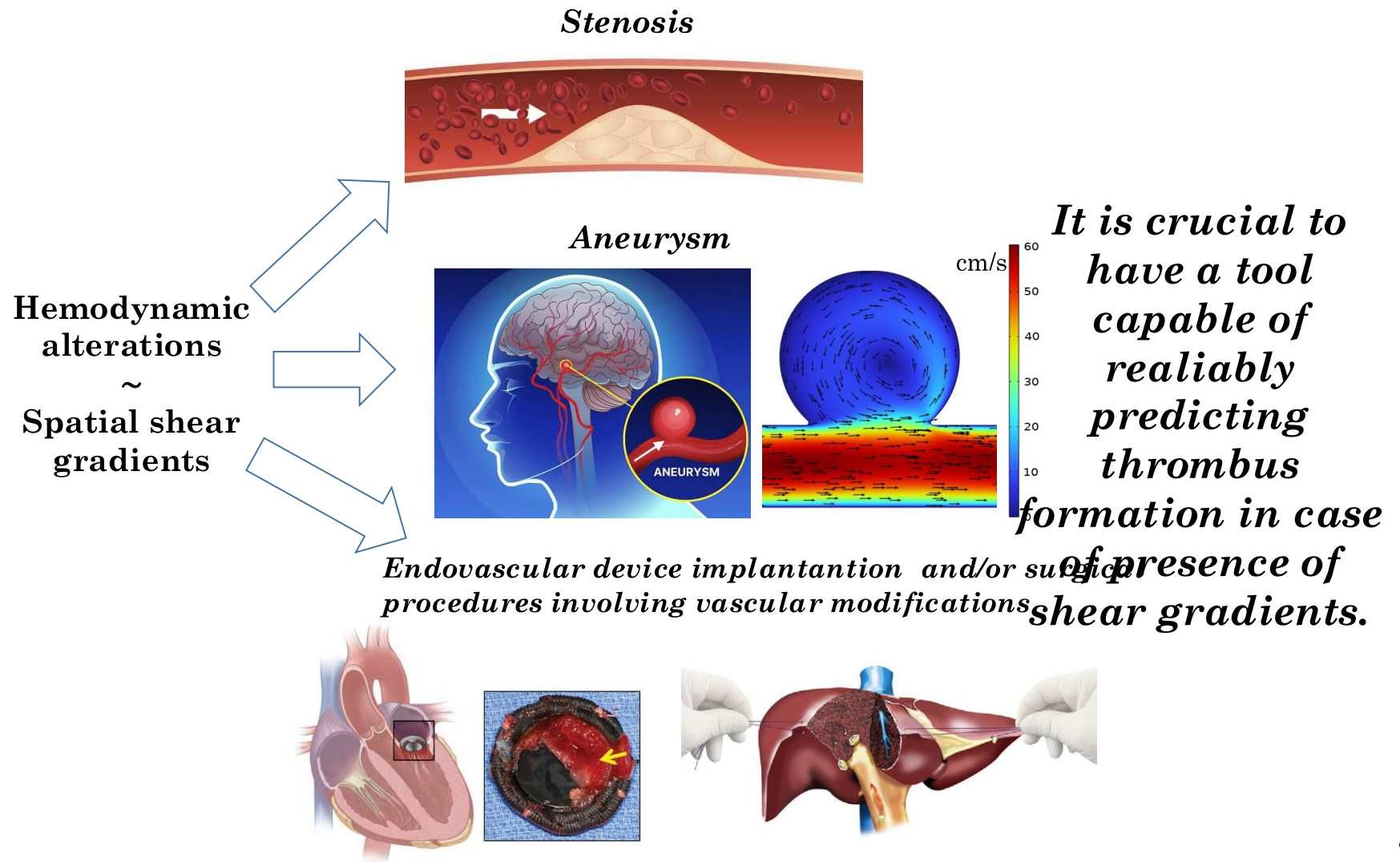
In vitro



rombus formation is iterations).

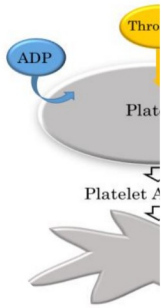


Introduction: Motivation behind our work



Features of thrombogenesis included in our model

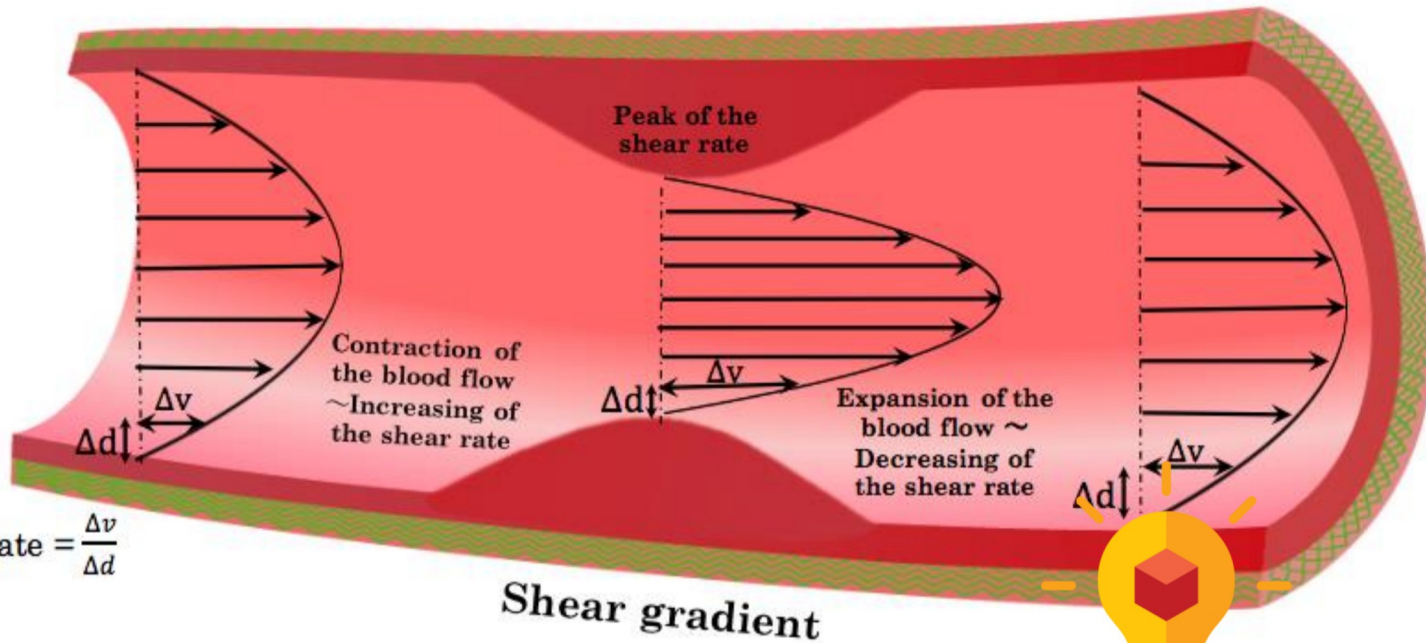
Chemical activation
Agonists



Mechanic
High shear



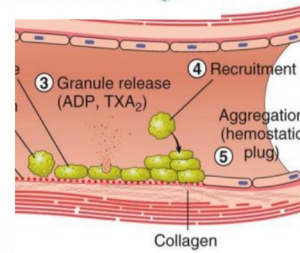
$$\text{Shear rate} = \frac{\Delta v}{\Delta d}$$



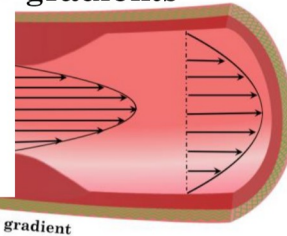
COMSOL
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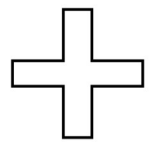
on
y



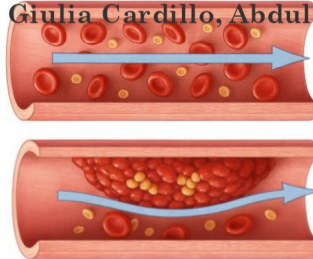
al aggregation:
gradients



A computational model of chemically and mechanically induced platelet plug formation
Giulia Cardillo, Abdul I. Barakat, BMMB, 2025

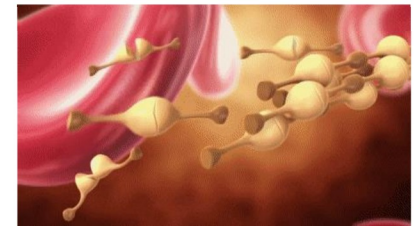


Influence of the
thrombus growth on
the blood flow field



&

Fibrin fibers
generation and
their stabilization
role



Modeling *chemical* platelet activation and aggregation

Hypothesis (Sorensen et al. 1999)



Platelets are
view as
soluble
biochemical
species

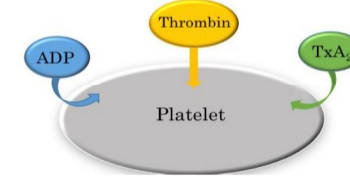
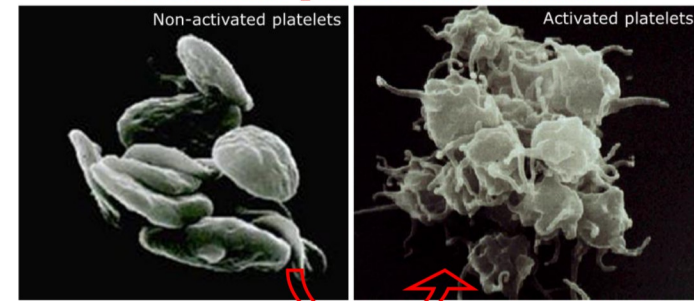
Set of CDR Equations (Sorensen et al. 1999)

$$\frac{\partial [C_i]}{\partial t} + \mathbf{div}(u \cdot [C_i]) = \mathbf{div}(D_i \cdot \mathbf{grad}[C_i]) + S_i$$

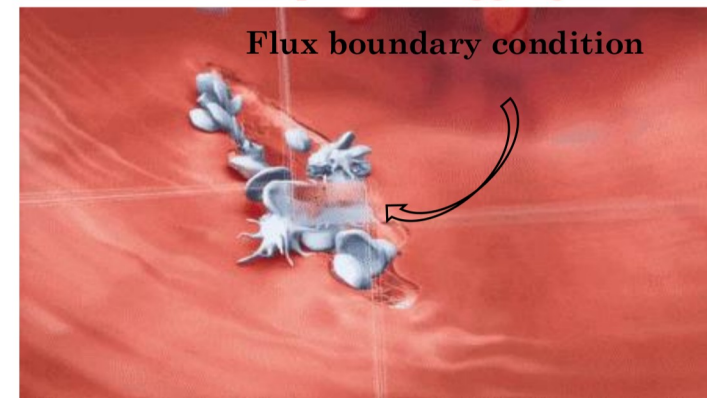
1. Resting platelets
2. Activated platelets
3. ADP
4. TxA2
5. Prothrombin
6. Thrombin
7. Fibrinogen
8. Fibrin
9. Antithrombin $\longrightarrow$ Anticoagulant agent

Agonists,
Coagulant agents

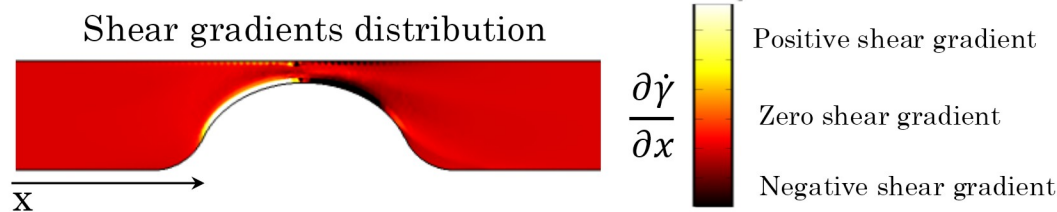
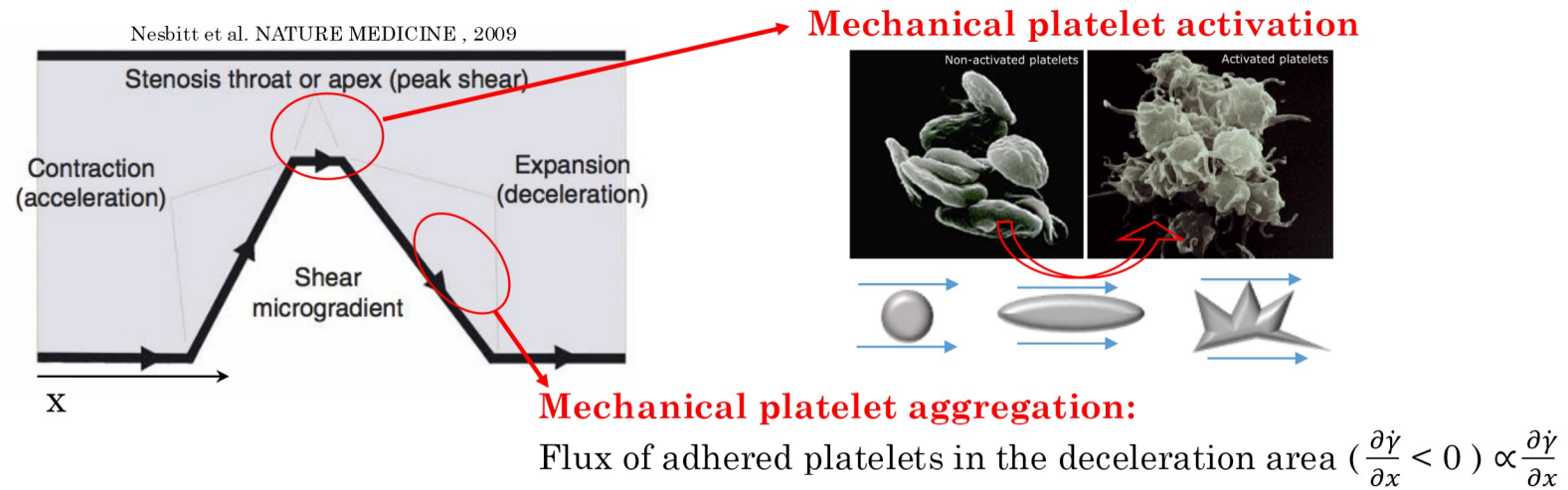
Chemical platelet activation



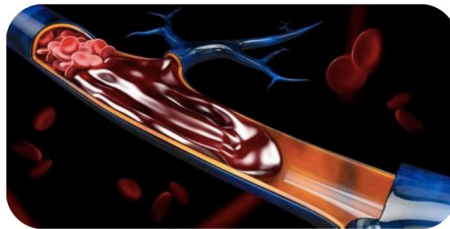
Chemical platelet aggregation



Modeling *mechanical* platelet activation and aggregation

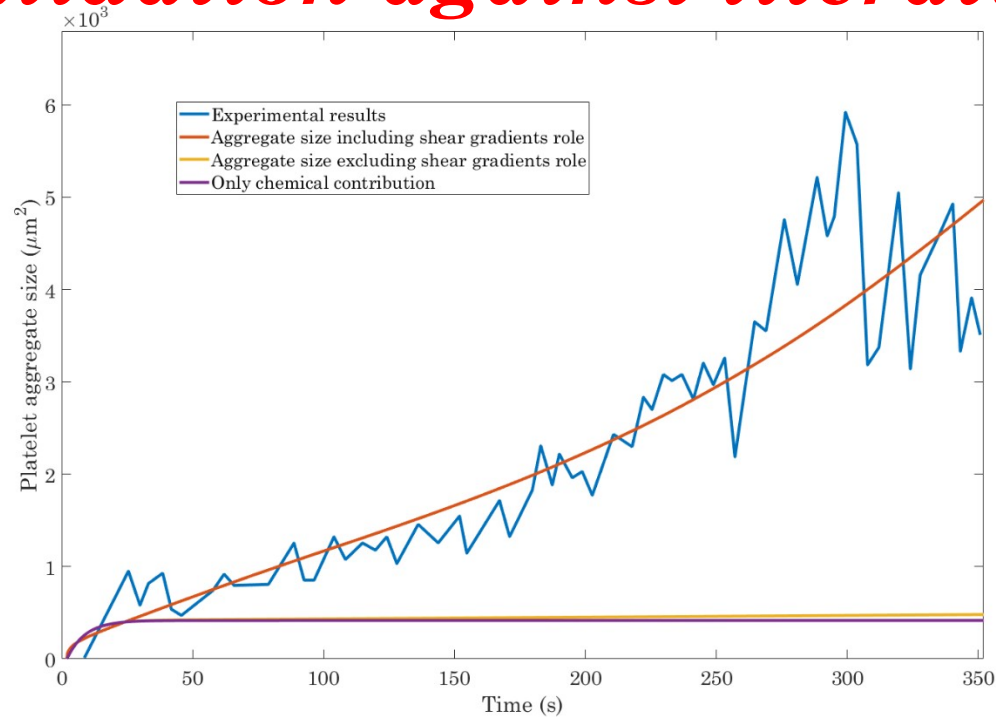


Modeling the *influence of the thrombus growth* on the flow field



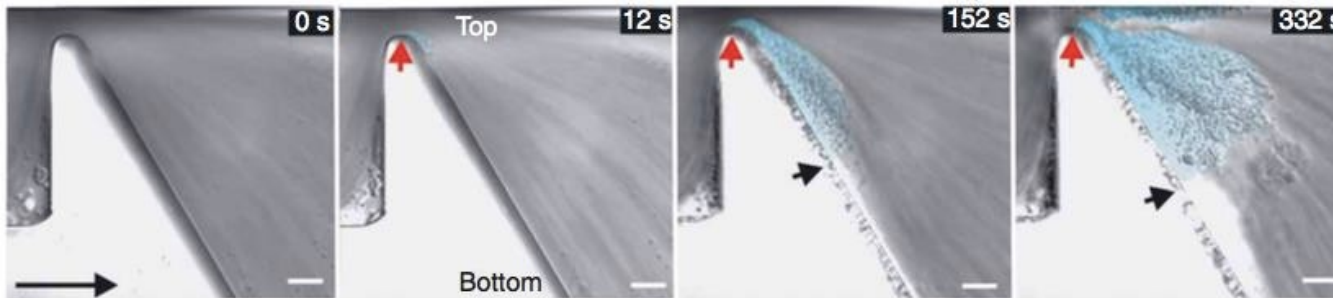
Thrombus is computationally modeled as a huge local increase of the blood viscosity.

Model validation against literature data



Cardillo & Barakat, 2025

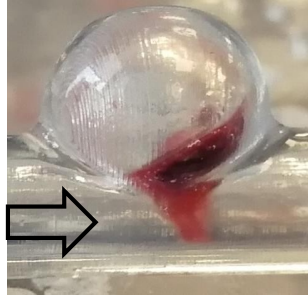
In-vitro thrombus formation in a micro-channel characterized by 90% stenosis



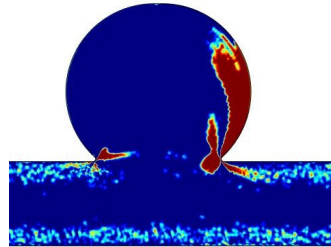
Nesbitt et al. NATURE
MEDICINE 2009

Model application in cerebral aneurysms

In-vitro thrombus formation

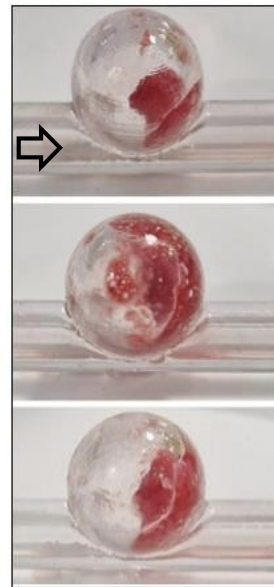
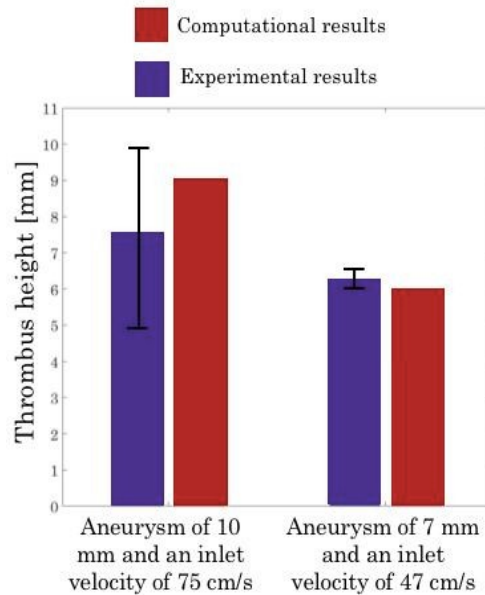


Areas of negative shear gradients

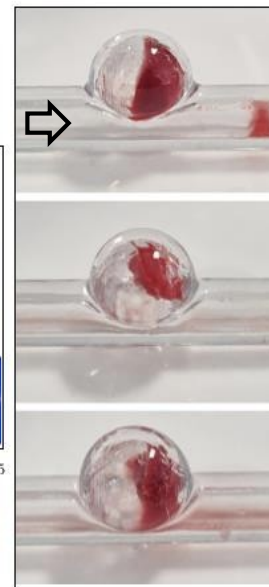
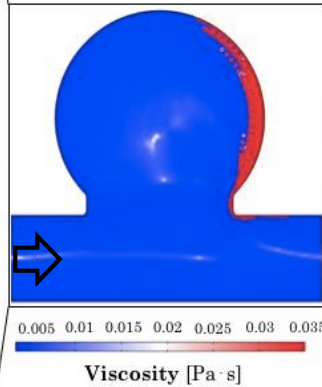


Red= areas of negative shear gradients

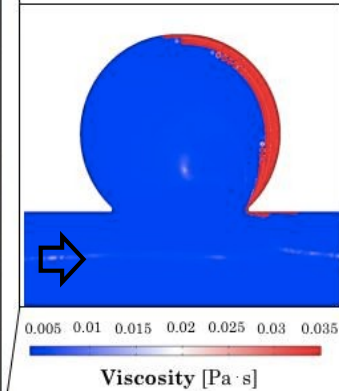
Pierre Pouponneau, PhD



10 mm-diameter aneurysm and an inlet velocity of 75 cm/s



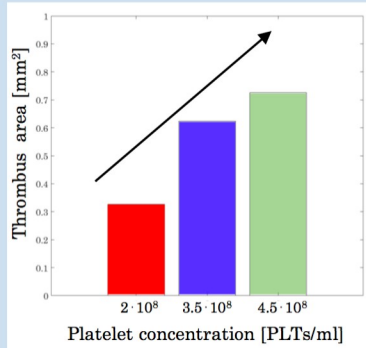
7 mm-diameter aneurysm and an inlet velocity of 47 cm/s



Model application in cerebral aneurysms

Effect on thrombosis of various patient-specific parameters

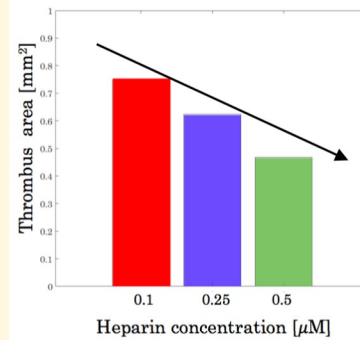
↑ Platelets count



↑ Thrombosis



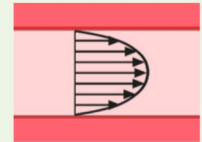
↑ Heparin



↓ Thrombosis



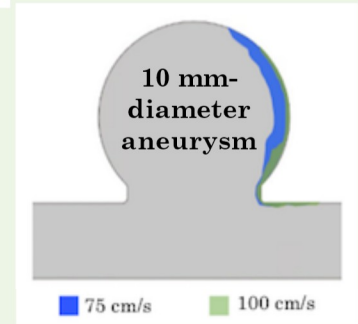
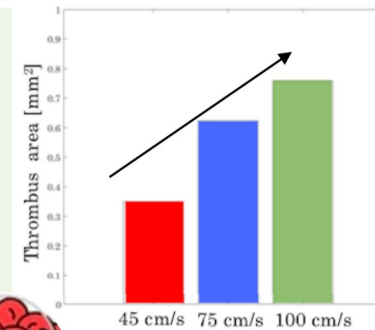
↑ Velocity in the parent artery



Short time scale:



Long time scale:



↑ Thrombosis



↓ Thrombosis



Pulsatility

Slight effect on thrombosis

Future direction: clinical application

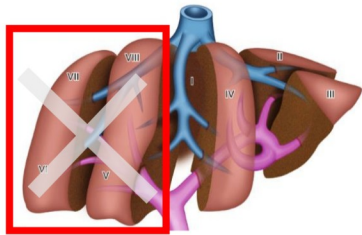
- Apply the model to patient specific geometries

Inria

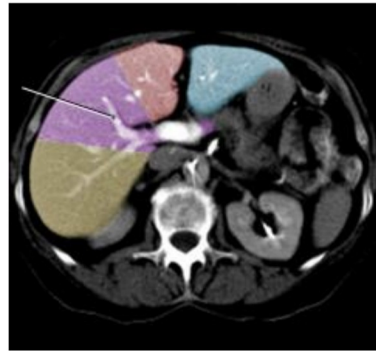
Irene
Vignon-Clementel



Hepatectomy



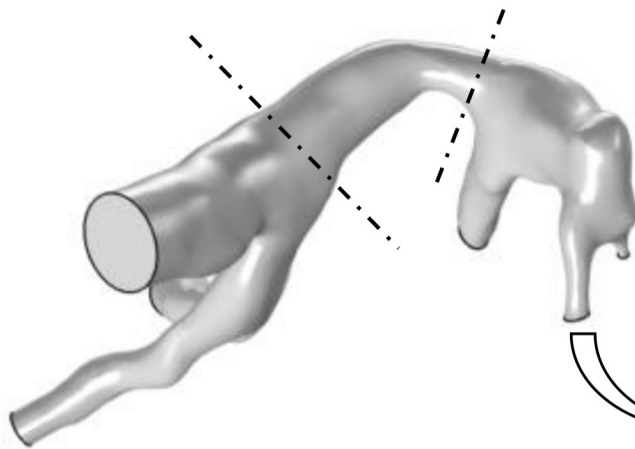
CT scan data



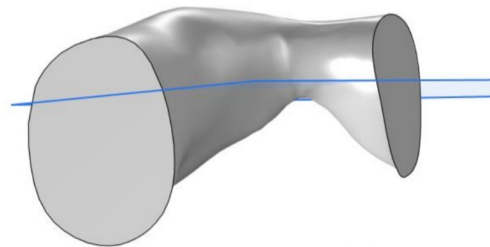
Simplified reconstruction of the
thrombus clinically observed



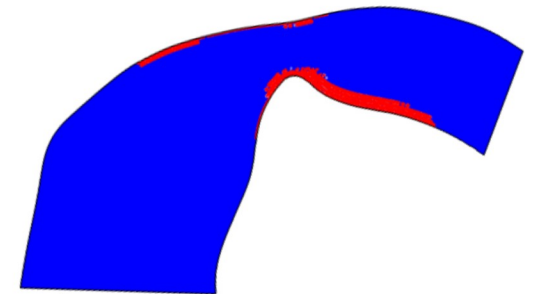
Patient specific 3D geometry



2D section geometry



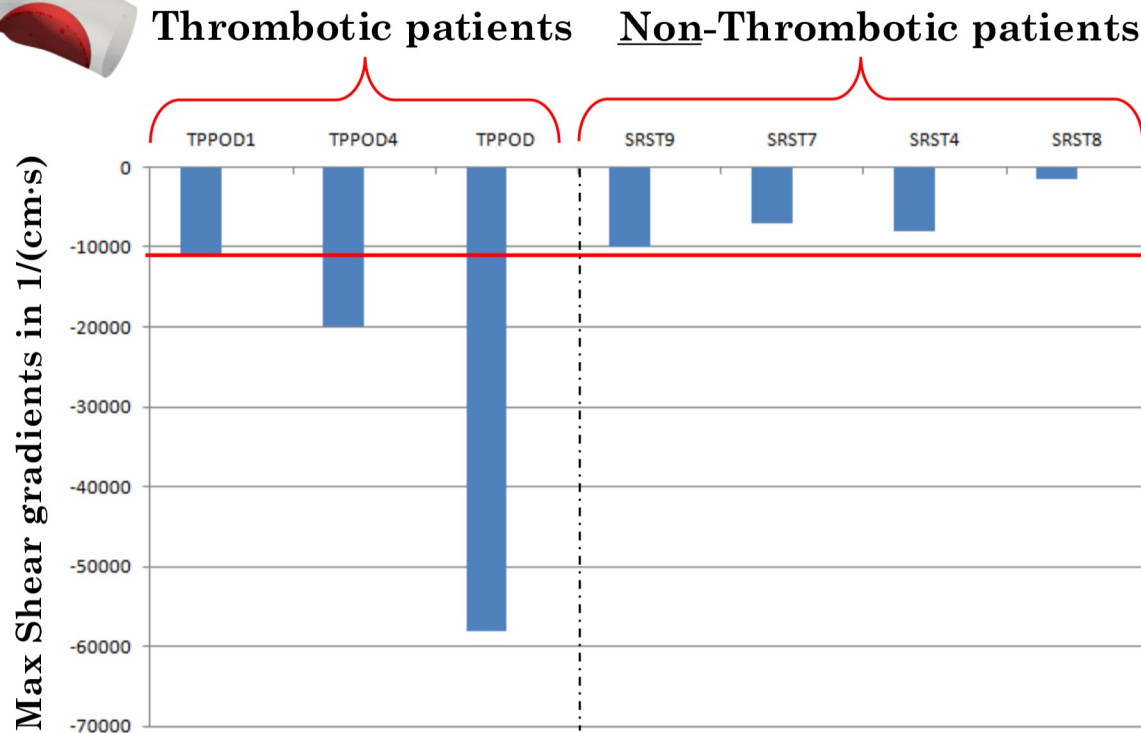
Thrombus computationally
predicted



Future direction: clinical application

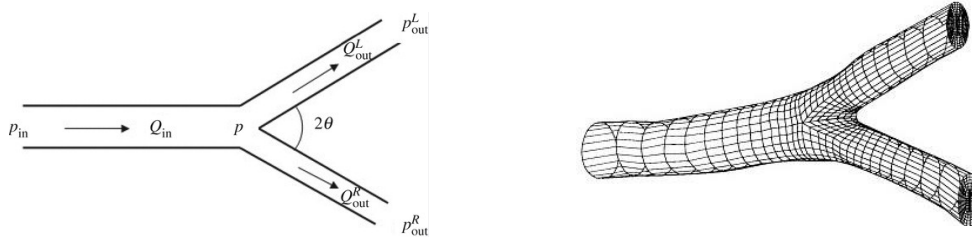


Negative shear gradients in thrombotic patients and non-thrombotic patients after hepatectomy

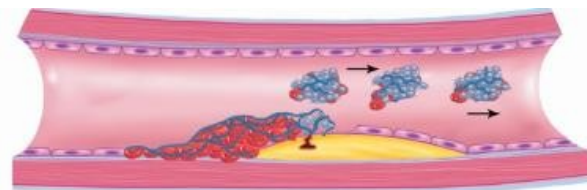
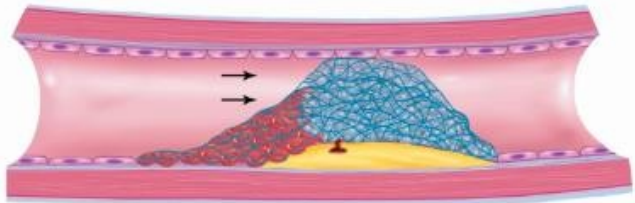


Take home messages and conclusions (1/2)

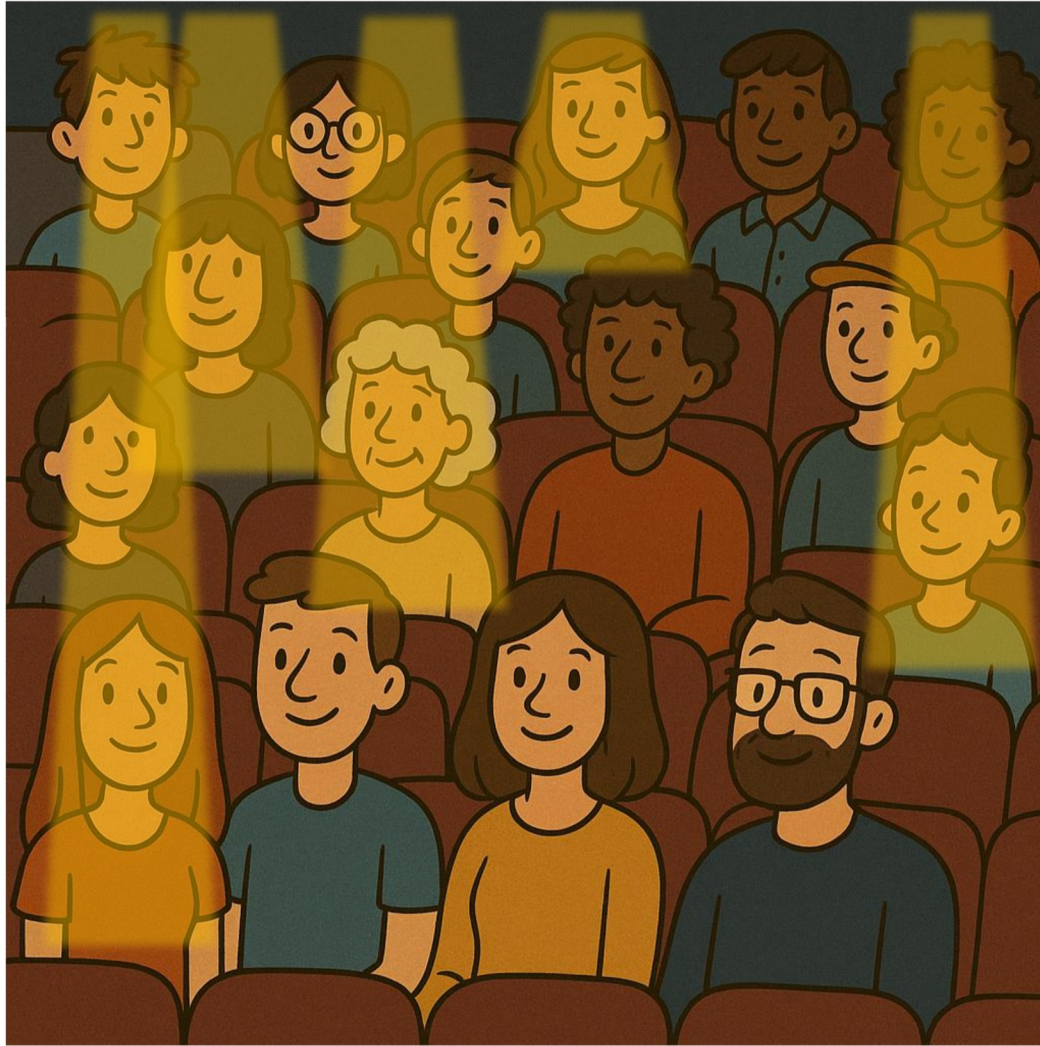
- In the proposed thrombus formation model, for the first time, the fundamental role of spatial shear gradients was included in addition to the role of biochemical factors.
- The computational model was validated and applied to different environments.
- Despite providing promising results, the developed model stands to benefit from different improvements. We are in the process of:
 - Extending the 2D proposed model to a 3D formulation



- Including the possibility of thrombus disaggregation



Take home messages and conclusions (2/2)



Thank you for your attention

Acknowledgements: LadHyx research group



AXA
Research Fund

Computational set-up

Inlet boundary conditions:

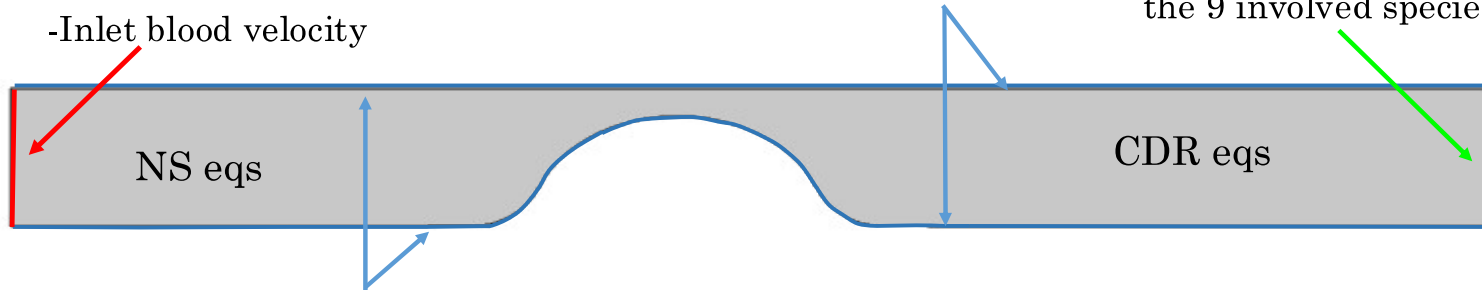
- Inlet concentration for the 7 species involved
- Inlet blood velocity

Fluid dynamics:

No slip condition

Outlet boundary conditions:

- Zero pressure outlet for the blood flow
- Outflow conditions for the 9 involved species



Platelet aggregation and species interactions:

- Mechanical flux boundary conditions imposed at all the walls but operates only in the portions of negative shear gradients (if formulation)
- Chemical flux boundary conditions in the portion of the injured endothelium

$$j_{r,mech}(x,t) = \begin{cases} 0, & \frac{\partial \gamma_w}{\partial x} \geq 0, \\ \frac{L}{\gamma_w} \left| \frac{\partial \gamma_w}{\partial x} \right| (S(x,t) \cdot k_{rs}[RP]), & \frac{\partial \gamma_w}{\partial x} < 0 \end{cases}$$

$$j_r(x,t) = S(x,t) \cdot k_{rs} \cdot [RP];$$

Coupled with the ODE system

Chemical platelet activation

$$k_{pa} = \begin{cases} 0, & \Omega < 1.0, \\ \frac{\Omega}{t_{act}}, & \Omega \geq 1.0, \end{cases} \quad \Omega = \frac{T}{T_{crit}} + \frac{APR}{APR_{crit}} + \frac{APS}{APS_{crit}}$$

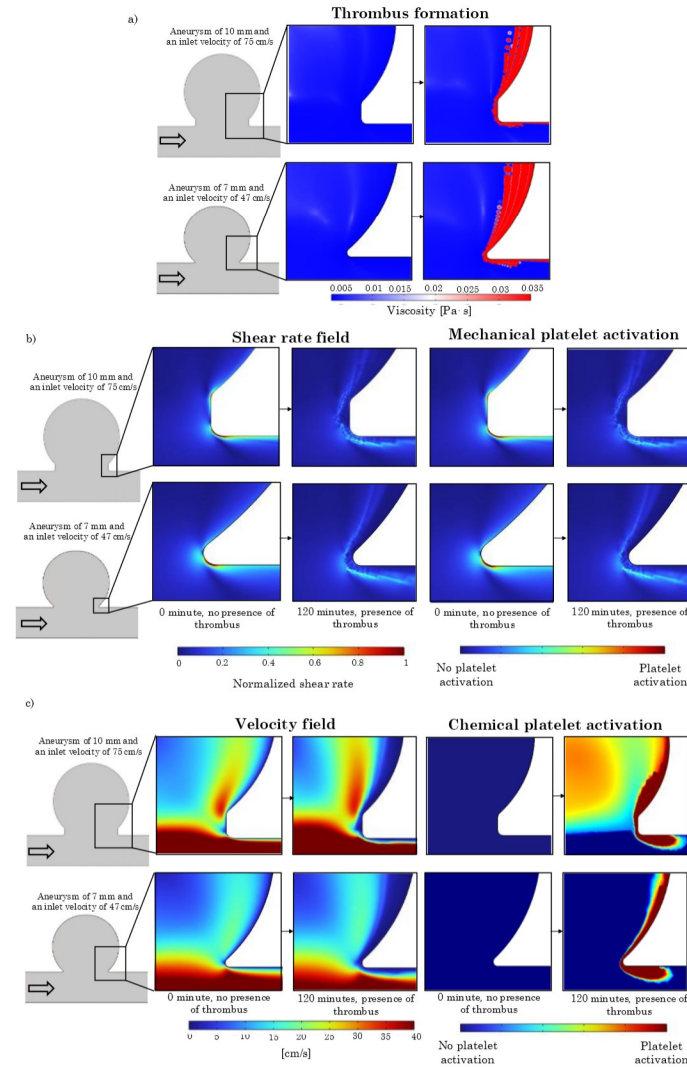
Mechanical platelet activation

$$k_{pa,mech} = \begin{cases} 0, & \frac{\dot{\gamma}}{\gamma_{crit}} < 1, \\ \frac{\dot{\gamma}}{\gamma_{crit} \cdot t_{act}}, & \frac{\dot{\gamma}}{\gamma_{crit}} \geq 1 \end{cases}$$

Figure 5: Influence of the thrombus formation on the flow field and consequences in platelet activation

Comparison of the viscosity with and without thrombus presence in the distal neck region, panel a. Blood flow field in absence of thrombus and in presence of the thrombus formed in 120 minutes in terms of velocity and consequences in the chemical platelet activation.

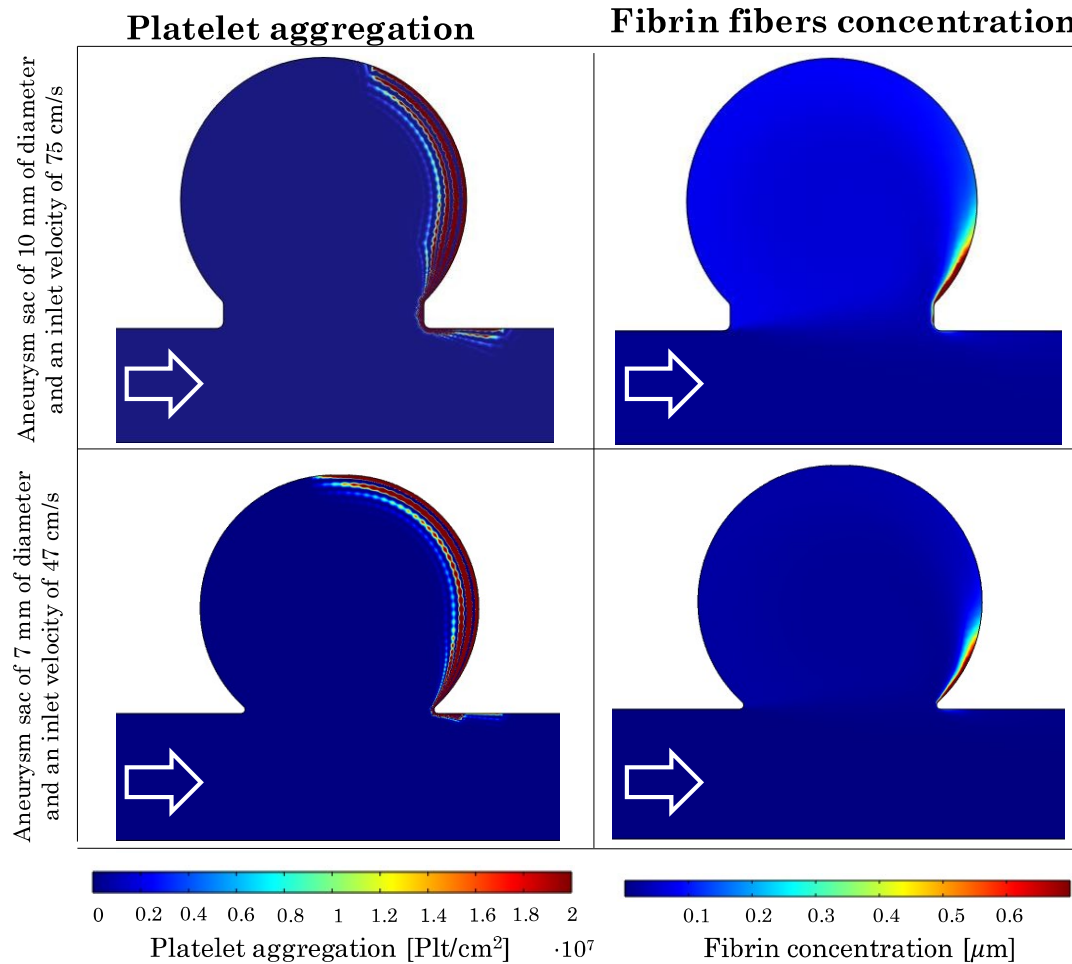
Blood flow field in absence of thrombus and in presence of the thrombus formed in 120 minutes in terms of shear rates and consequences in the mechanical platelet activation.



Key message:

-Shear rate: when there is no thrombus formation, the critical value at which the mechanical platelet activation is observed, is reached at the distal part of the aneurysm's neck, where the fluid vein impacts against the distal neck wall. The presence of the thrombus reduces the impact of the blood flow against the distal neck wall and, as a consequence, the shear rate is reduced and the critical value is not reached. This results in an initial mechanical platelet activation, for both the configurations, localized at the distal neck that reduces along the time.

-Chemical activation: reached where we have regions of low velocity because there the coagulant agents have time to accumulate



Key messages:

- Trigger of the thrombus formation is the platelet plug formation, but then the formation of the fibrin fiber stabilize the clot
- In both geometries, the thrombus formation is primarily the result of platelet aggregation. Fibrin concentration reaches its critical value in the distal part of the aneurysm's sacs. In these regions, the thrombus is more stable, it is composed of aggregated platelets and fibrin fibers, this computationally translates in a blood viscosity two times higher than the regions in which the thrombus growth is only attributed to platelet aggregation.

Figure 7: Thrombus composition after 2 hours: Platelet aggregation and Fibrin concentration distribution

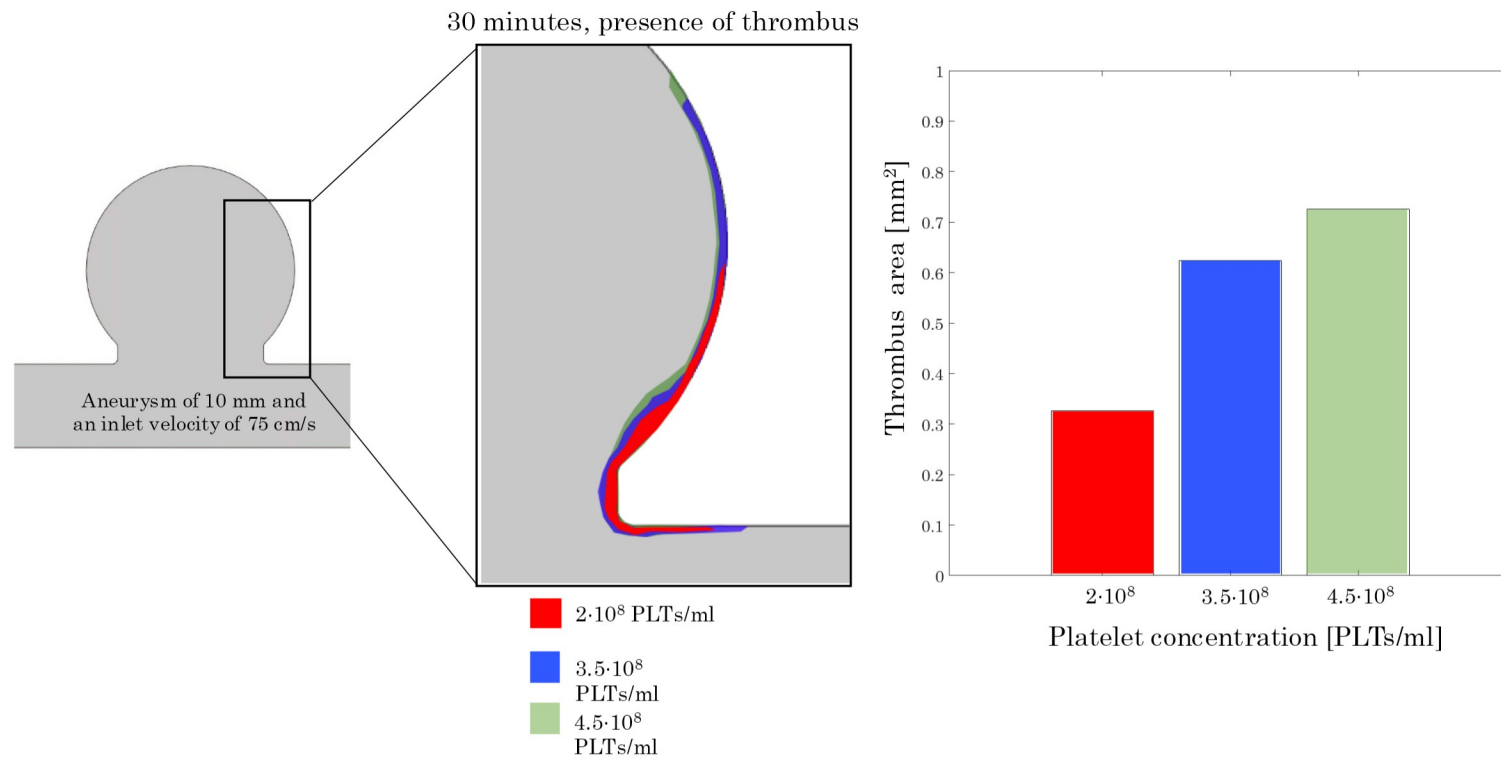


Figure 8: Sensitivity analysis: Role of the platelet count

Key messages:

- Platelet count greatly differs in different patient
- Higher platelet concentration in the blood induces greater thrombus aggregate size.
- A higher number of the platelets in the unit of blood volume results in a greater availability of platelets to aggregate and create the thrombus in the same amount of time.

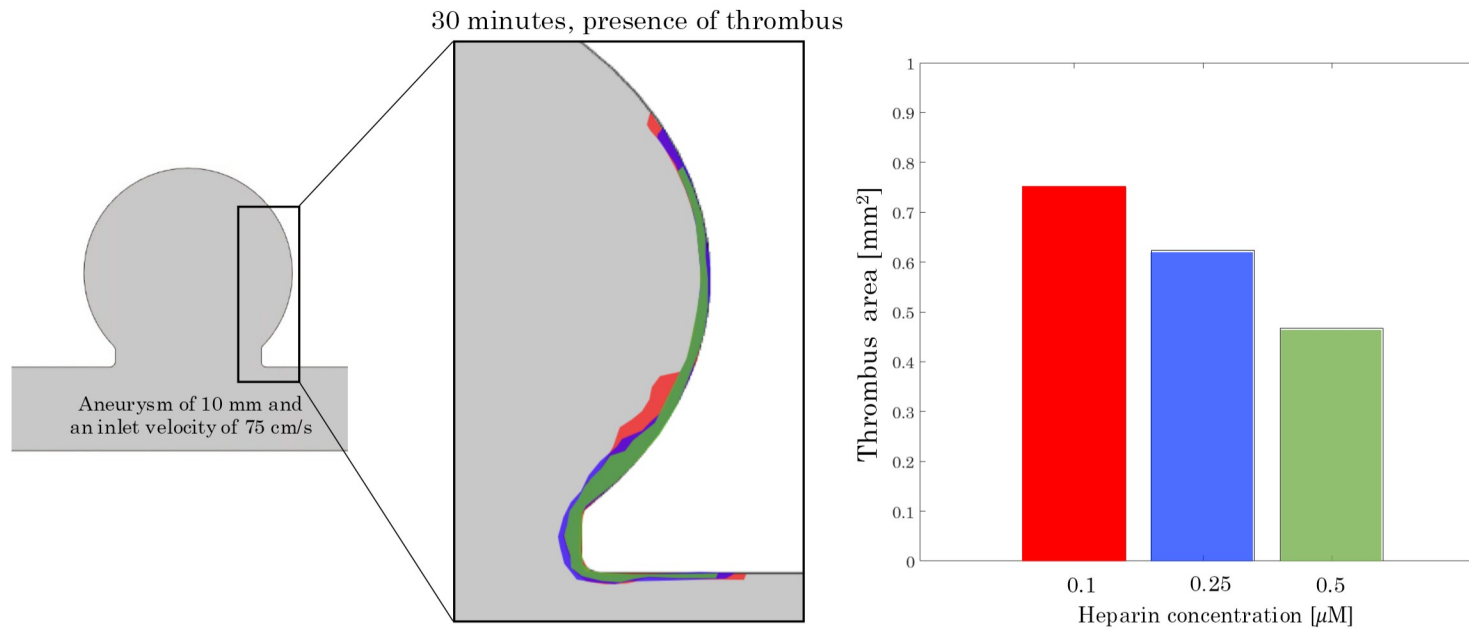


Figure 9: Sensitivity analysis: role of heparin concentration

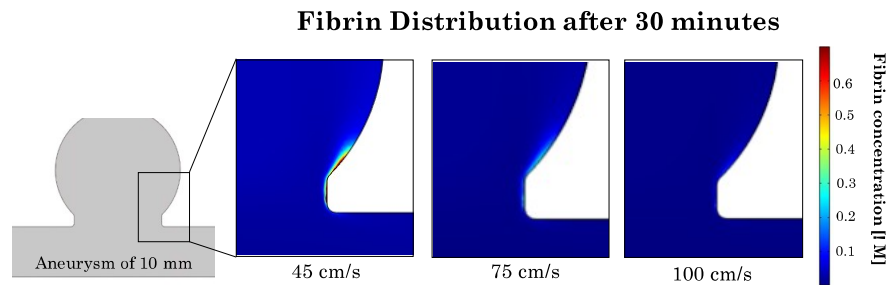
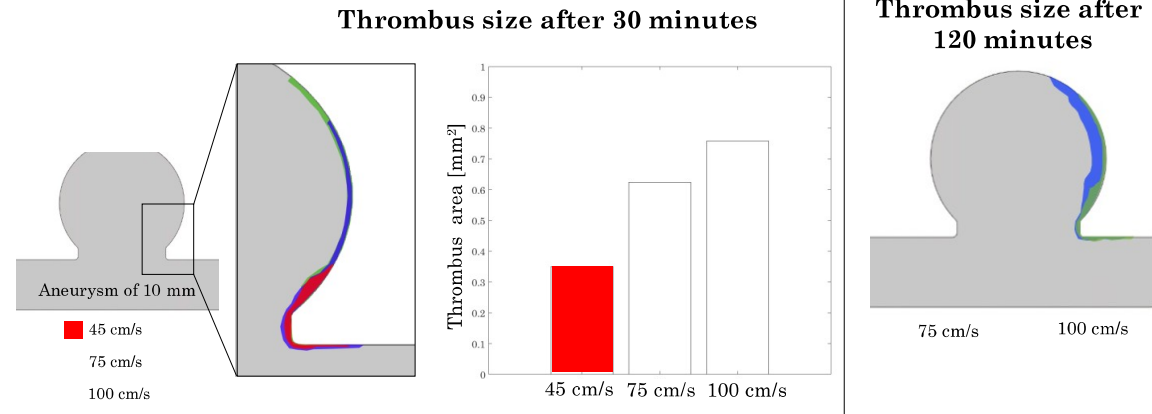
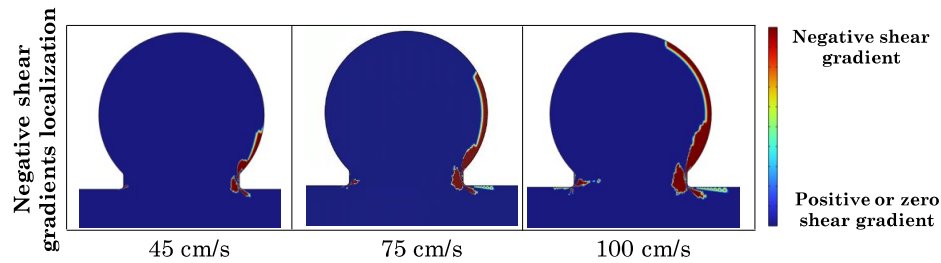
Key messages:

- Heparin is an anticoagulant agent and in this model, its role in inhibiting thrombin generation was taken into account.
- Heparin concentration and thrombus size are inversely related.
- An increase in the heparin concentration results in a decrease of thrombin generation that in turn leads to two consequences: a decrease in chemical platelet activation, which is proportional to the thrombin concentration, and a decrease in the fibrin concentration, the rate of conversion of fibrinogen in fibrin directly depends on thrombin formation.

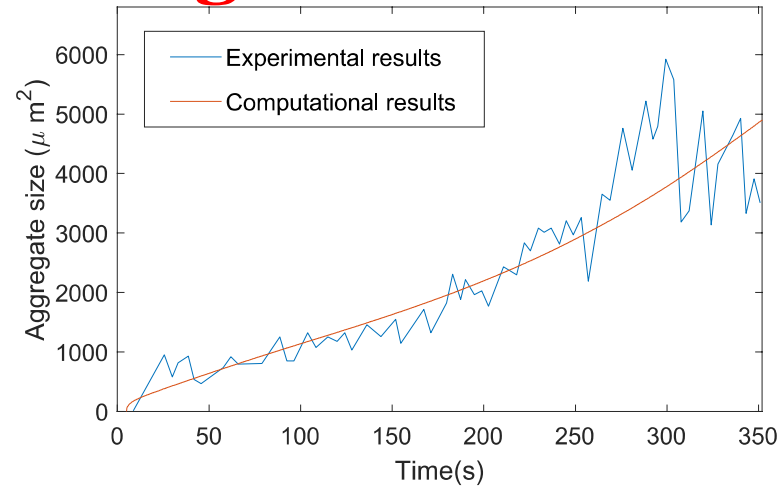
Figure 10: Sensitivity analysis: role of the inlet velocity in the thrombus formation, in the negative shear gradient regions and in fibrin fibers formation after 30 minutes

Key messages:

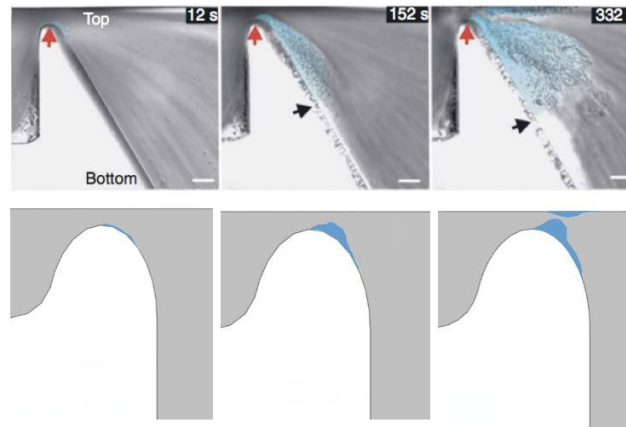
- At 30 minutes: higher inlet velocities induces greater thrombus aggregate size. This because higher velocities induces higher shear rate values and greater regions interested by NSG and, as a consequence a grater mechanical platelet activation and aggregation.
- Lower velocity induces higher fibrin generation process, more stable clot
- At longer time scale, the mechanical platelet activation becomes less important and we have an inversion of the trend



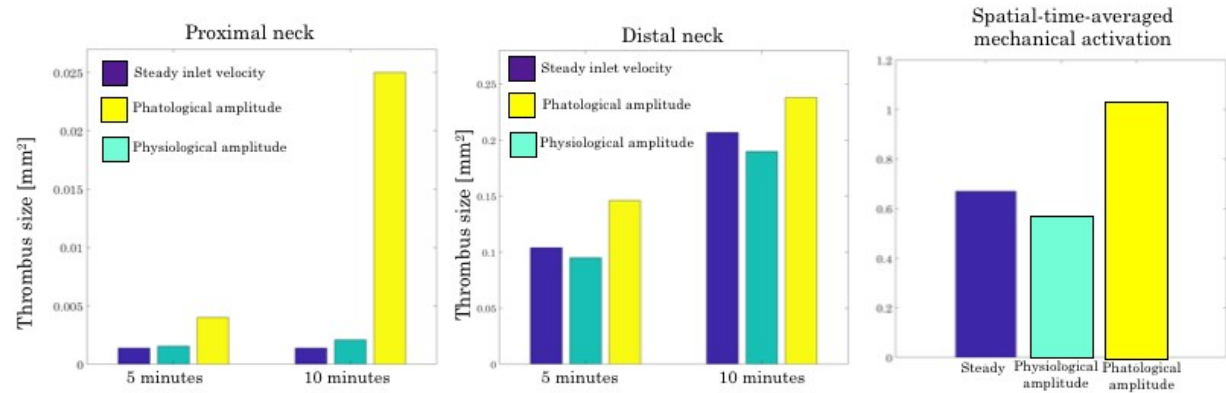
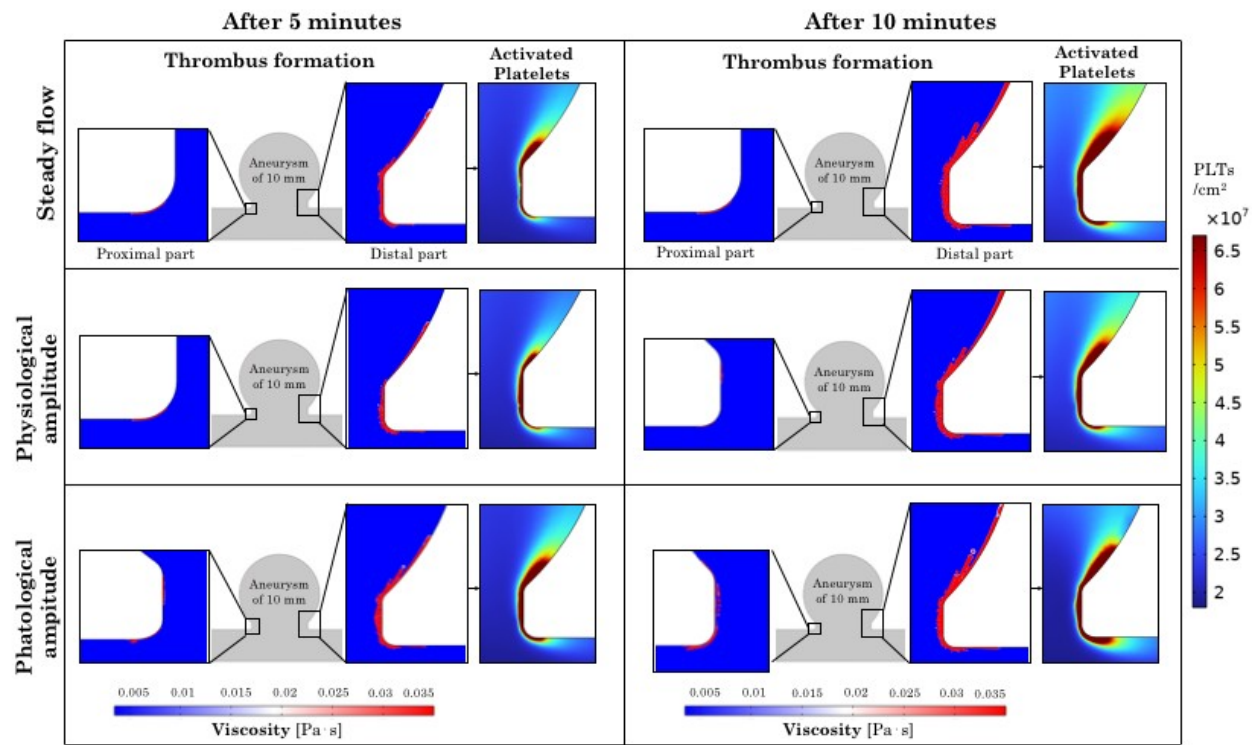
Model validation against literature data:



In-vitro thrombus formation in a micro-channel characterized by 90 % of stenosis



Nesbitt et al. NATURE MEDICINE Volume 15 Number 6 JUNE 2009



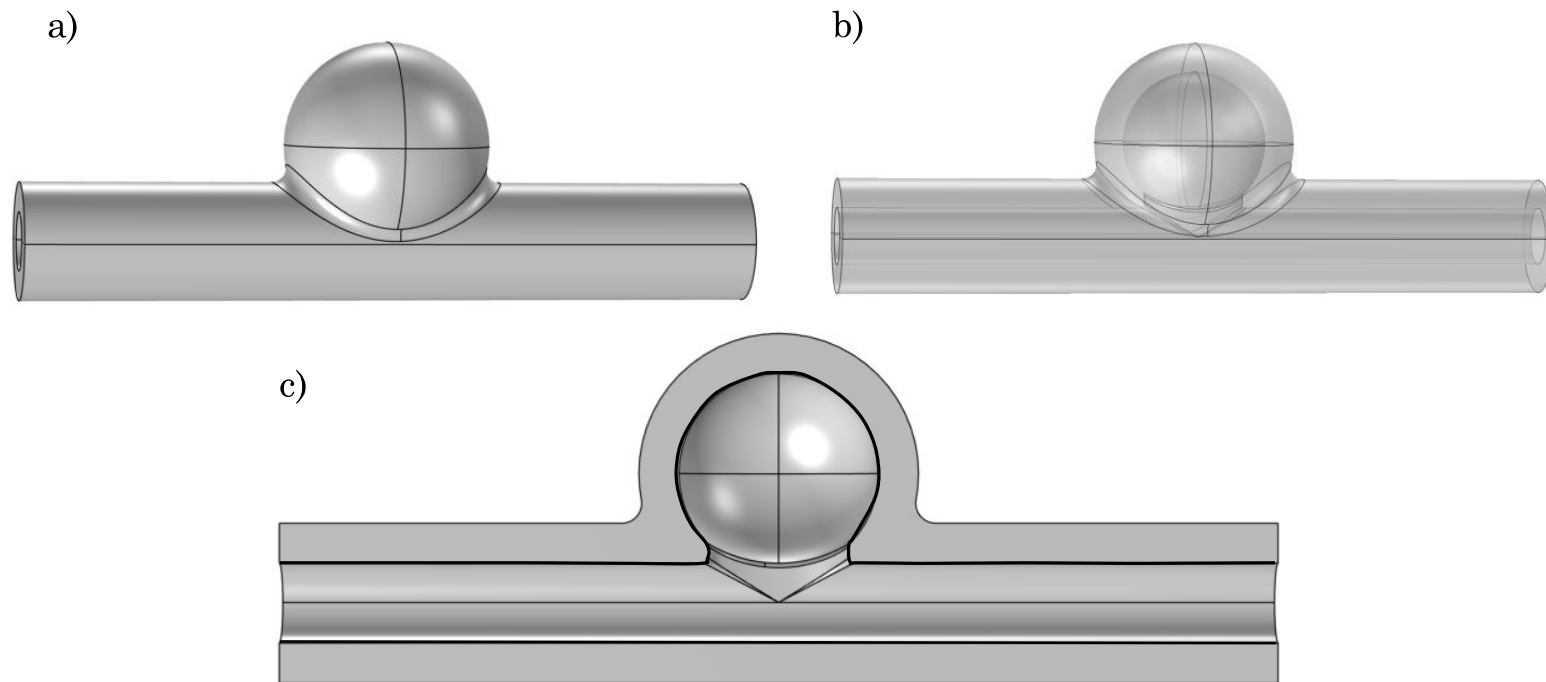
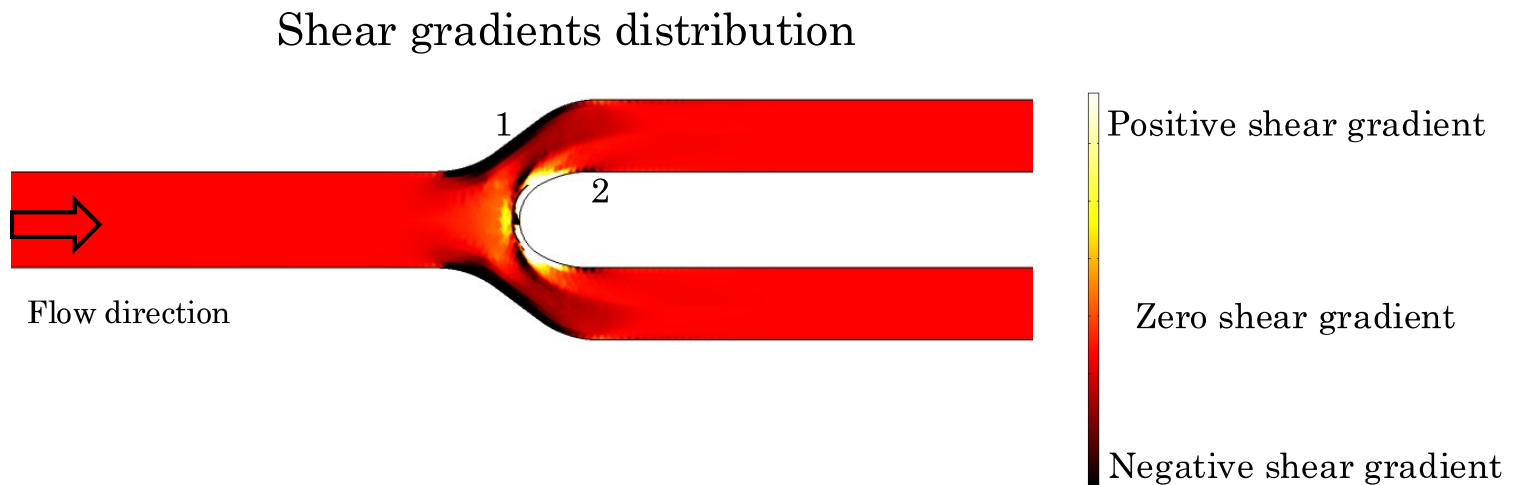
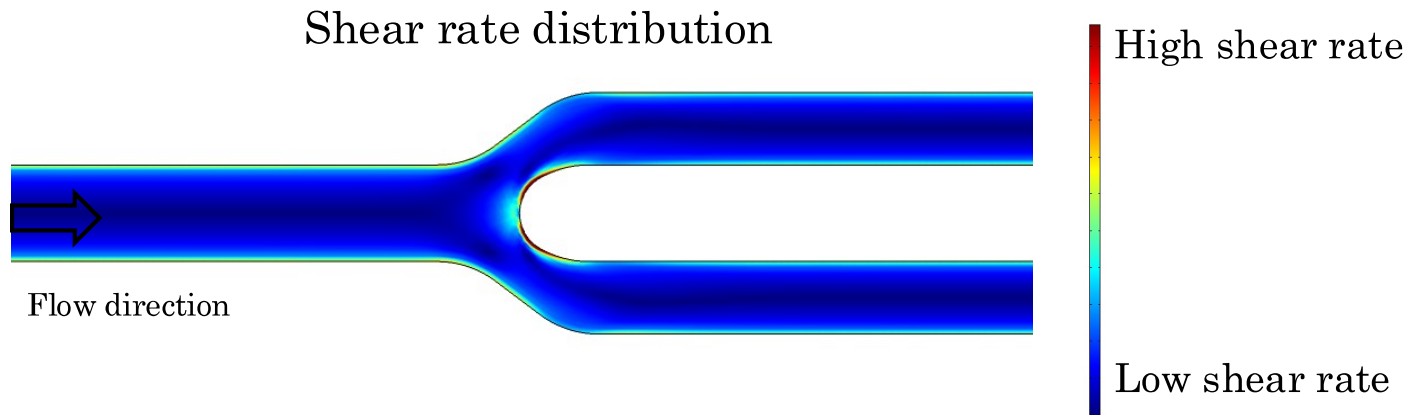
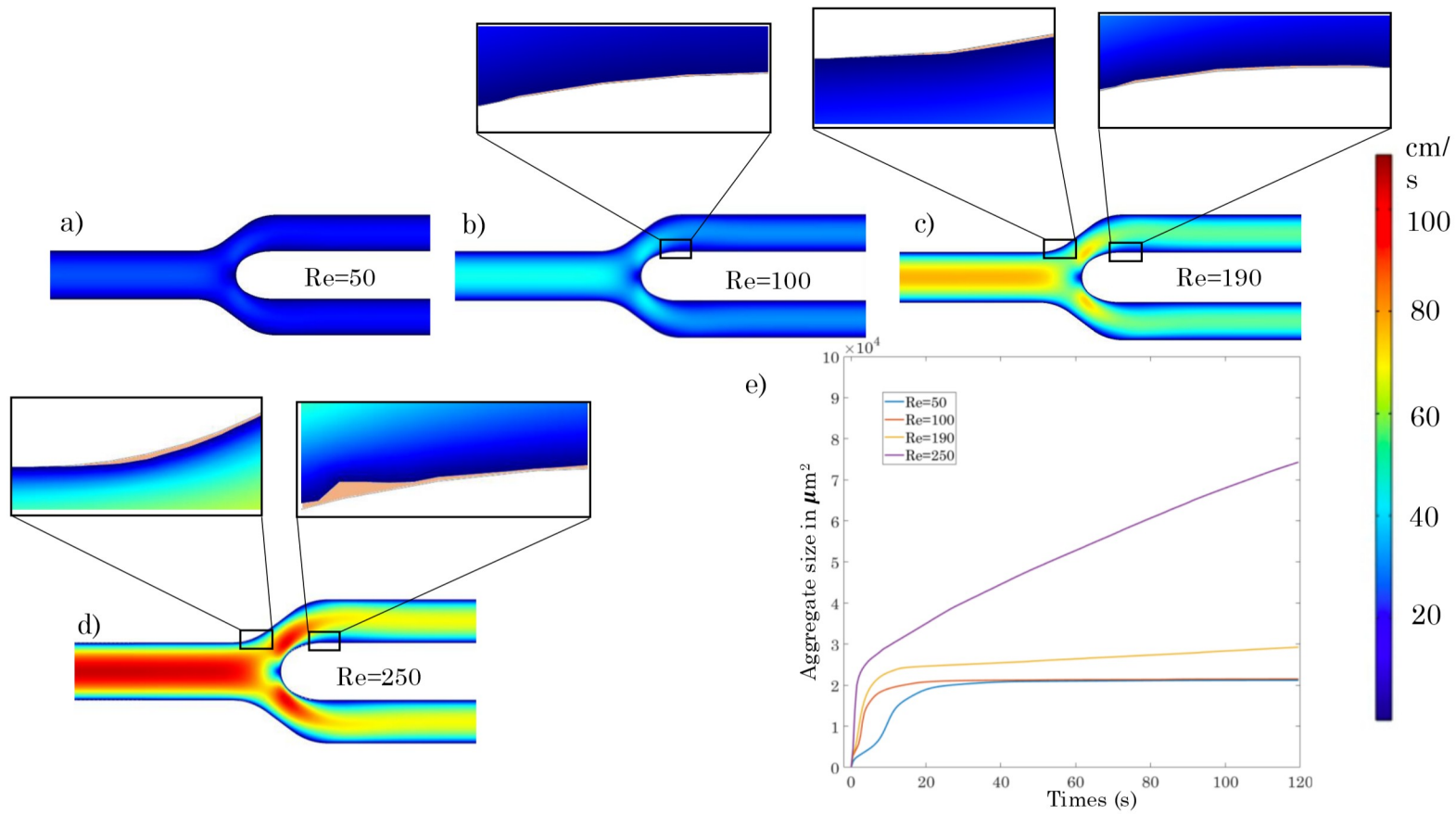


Figure : Idealized geometry of an aneurysm with a diameter of 10 mm. a) External 3D domain b) Transparent 3D domain c) Longitudinal section

Results: Shear rate and shear gradients in a bifurcation





Some important parameters of the model

Name	Value	Description
D_{plt}	$1.58 \cdot 10^{-6} \text{ [cm}^2/\text{s]}$	Platelet diffusion coefficient
D_{ADP}	$2.57 \cdot 10^{-6} \text{ [cm}^2/\text{s]}$	ADP diffusion coefficient
D_{T}	$4.16 \cdot 10^{-7} \text{ [cm}^2/\text{s]}$	Thrombin diffusion coefficient
D_{TxA2}	$2.14 \cdot 10^{-6} \text{ [cm}^2/\text{s]}$	TxA2 diffusion coefficient
D_{PT}	$3.32 \cdot 10^{-7} \text{ [cm}^2/\text{s]}$	Prothrombin diffusion coefficient
D_{AT}	$3.49 \cdot 10^{-7} \text{ [cm}^2/\text{s]}$	Anti-thrombin diffusion coefficient
T_{crit}	0.05 [U/ml]	Thrombin Activation treshold
ADP_{crit}	$2 \text{ [}\mu\text{M]}$	ADP Activation treshold
$\text{TxA2}_{\text{crit}}$	$0.6 \text{ [}\mu\text{M]}$	TxA2 Activation treshold
$\dot{\gamma}_{\text{crit}}$	10000 [1/s]	Shear rate activation theshold
V_{plt}	$4.18 \cdot 10^{-12} \text{ [cm}^3\text{]}$	Volume of a platelet
k_{rs}	0.0037 [cm/s]	Rate of adhesion of resting platelets
$k_{\text{as}}=k_{\text{aa}}$	0.135 [cm/s]	Rate of adhesion of activated platelets

Parameters of the Sorensen model

Species	$D_{b,i}$ (cm ² s ⁻¹) ^a	$k_{1,j}$ (s ⁻¹) ^b	λ_j (nmol PLT ⁻¹) ^c	$s_{p,j}$ (nmol PLT ⁻¹ s ⁻¹) ^d	Initial conc. in blood	$[a_{j,crit}]^e$
Resting PLT	1.58×10^{-9}	N/A	N/A	N/A	$1.5 \times 10^8 - 3 \times 10^8$ PLT ml ⁻¹	N/A
Activated PLT	1.58×10^{-9}	N/A	N/A	N/A	$0.05 \cdot [RP]$	N/A
ADP	2.57×10^{-6}	N/A	2.4×10^{-8}	N/A	0.00 μ M	2.00 μ M
TxA ₂	2.14×10^{-6}	0.0161	N/A	9.5×10^{-12}	0.00 μ M	0.60 μ M
Prothrombin	3.32×10^{-7}	N/A	N/A	N/A	1.100 μ M	N/A
Thrombin	4.16×10^{-7}	N/A	N/A	N/A	0.00 U	0.10 U ml ⁻¹
ATIII	3.49×10^{-7}	N/A	N/A	N/A	2.844 μ M	N/A

α	1.00	...
β	9.11×10^{-3}	nmol U ⁻¹
M_∞	7.00×10^6	PLT cm ⁻²
ϕ_{at}	3.69×10^{-9}	U PLT ⁻¹ s ⁻¹ (μ M PT) ⁻¹
ϕ_{rt}	6.50×10^{-10}	U PLT ⁻¹ s ⁻¹ (μ M PT) ⁻¹
μ	0.035	g cm ⁻¹ s ⁻¹
ρ	1.1064	g cm ⁻³